

Every funkopop you see in the background of an anti-ai influencer requires as much water to produce as 200 years of prompting an AI

Posted by u/UwU_Spank_Me_Daddy • 0 points • 36 comments

Now that we have actual measurements for energy and water usage of AI datacenters, I thought it would be fun to compare them to the production of a funkopop, cos a lot of anti-ai influencers seem to have more and more build up in the background whenever they have some new pseudery to preach.

Turns out a funkopop requires 11,019x more energy and 73,000x more water per prompt. It would take you 200 years of prompting every day to reach the same water usage of 1 funkopop and 30 years for energy.

A standard funkopop weighs 170g, and a super size weighs in at around 283g. We'll ignore the super size. The inks take up some of the mass, and maybe they don't contain water, so to be conservative and make the math easier to follow, let's just call it 100g.

Suspension-PVC manufacturing uses 77.2 MJ of embodied energy per kg of PVC.

$77.2\text{MJ/kg} \times 0.1\text{kg} = 7.72\text{MJ}$. Converting that to watt-hours: $7.72\text{MJ} \times (1\text{kWh}/3.6\text{MJ}) = 2.144\text{kWh}$

MDPI says the injection molding process requires 5 kWh/kg. For 0.10 kg, this would be 0.50 kWh, which brings it to 2.644 kWh.

According to Rwsci, PVC production requires 180L of water per kg. [watercalculator.org] (<http://watercalculator.org>) says 290L. We'll take the conservative estimate. $180\text{L/kg} \times 0.1\text{kg} = 18\text{L}$

Adding insult to injury, according to the Ecolnvent Dataset, for every kg of PVC, factories require 0.011 cubic meters of water for cooling. So $0.011\text{ m}^3 = 11\text{ L}$ (since $1\text{ m}^3 = 1,000\text{ L}$)
For a 0.1 kg Funko Pop: $11\text{L/kg} \times 0.1\text{kg} = 1.1\text{L}$.

Total: 19.1L

I had to use log scale for the graph because otherwise the bars for gemini prompts get rounded down to 0 pixels.

For energy usage, the assumption here is that the Co2 emissions of plastic production from fossil fuels are the same as an AWS/Azure datacenter that runs on renewables.

Embodied energy for PVC production:

[https://en.wikipedia.org/wiki/Embodied_energy#In_common_materials]

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Injection molding energy for PVC forming: [<https://www.mdpi.com/2073-4360/16/21/3097>]

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PVC production water requirements:

[[https://responsiblewaterscientists.wordpress.com/2017/06/16/the-water-footprint-of-](https://responsiblewaterscientists.wordpress.com/2017/06/16/the-water-footprint-of-plastics/)

[plastics/](https://responsiblewaterscientists.wordpress.com/2017/06/16/the-water-footprint-of-plastics/)](<https://responsiblewaterscientists.wordpress.com/2017/06/16/the-water-footprint-of-plastics/>)

PVC cooling water requirements:

[<https://www.sciencedirect.com/science/article/abs/pii/S0959652615010501>]

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Energy and water usage per gemini prompt: [<https://arxiv.org/abs/2508.15734>]

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Comments

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u/[deleted] • 1 points • 2025-09-12 13:22 UTC

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u/UwU_Spank_Me_Daddy • 1 points • 2025-11-24 09:09 UTC

No, all the pre-requisites you mentioned were included in the paper.

Even the "existing approach" includes training and storage. Comprehensive approach, which was the figure I used, goes even further, tracking a bunch of minor things in the production chain that all add up.

[https://preview.redd.it/bkq8x93i763g1.png?](https://preview.redd.it/bkq8x93i763g1.png?width=1062&format=png&auto=webp&s=7fd3c0a1242e642babcc9b6d8a2b0c8d384466be)

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For the funk pop, I'm not including the entire production chain, I just took 2 major sources of water use required to cool the PVC, because that's all that was necessary

It seems like you're confused because you're used to seeing what anti-ai do. They look at the aggregate of 1 billion users on a datacenter and just say "wow that's a lot of water in total" as if to compare it to someone making a cup of coffee or something, even though the per capita usage is 0.26mL, and we regularly handle aggregate water usage like that every day.

u/[deleted] • 1 points • 2025-09-09 10:47 UTC

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u/[deleted] • 1 points • 2025-09-07 23:28 UTC

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u/Imperator-Solis • -5 points • 2025-09-07 20:23 UTC

Why are we measuring this in water now? the most abundant, easily obtainable resource in the universe?

u/honato • 19 points • 2025-09-07 20:41 UTC

Water usage is one of the common measurements used by antis.

u/foxtrotdeltazero • 2 points • 2025-09-07 22:37 UTC

they can measure things now!?

i sweated 7 waters just reading this comment

u/honato • 2 points • 2025-09-08 00:40 UTC

Just be glad (or sad) that you missed denamrks as a measuring tool.

u/hel-razor • 7 points • 2025-09-07 20:01 UTC

Now do Shein